

WCS Upper A-Arms Simulation Results

MadRams

Tecnológico de Monterrey

Campus Guadalajara

Forces estimation for WCS:

Total Car weight: $293 \text{ kg} = 2875 \text{ N}$.

Weight distribution: 40/60.

Bump + Turn (Only 2.5g lateral for lower arms WCS) = 4g vertical ; 1.5g lateral.

Rear Tires Forces : $F_x = 1293 \text{ N}$; $F_y = 3448 \text{ N}$

Front Tires Forces (Upper WCS) : $F_x = 862.5 \text{ N}$; $F_y = 2300 \text{ N}$

Front Tires Forces (Lower WCS) : $F_x = 1437.5 \text{ N}$; $F_y = 575$

FBD Front Arms:

FBD Rear Arms:

Front Upper A-Arms

Configuration 1:

This configuration represents the baseline design, utilizing an all-steel construction. The main tubular members are modeled in AISI 4130 Chromoly, while the mounting brackets (ears) are made of AISI 1020 Steel. This setup provides high global stiffness ($E = 200 \text{ GPa}$) and excellent yield strength, ensuring minimal deflection and a high Factor of Safety (FOS) under Worst-Case Scenario (WCS) loads. However, it represents the heaviest iteration of the A-Arm.

Configuration 2:

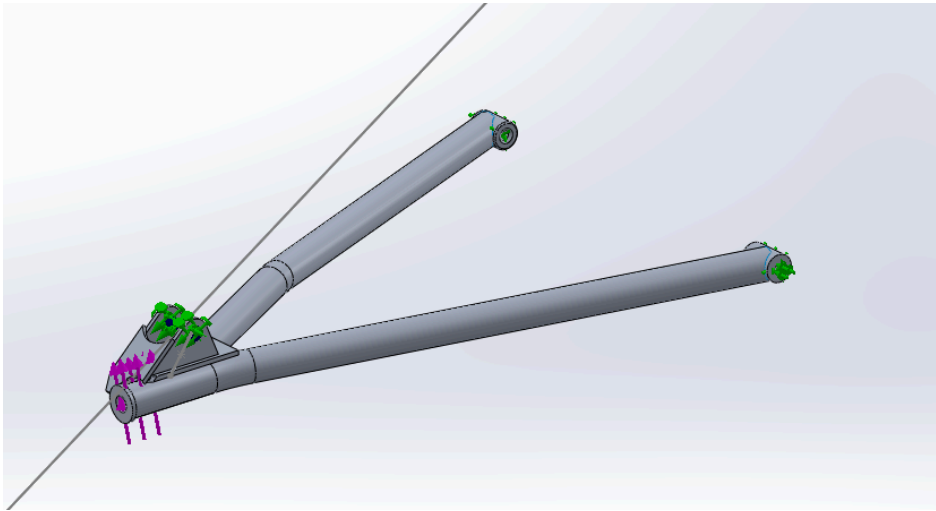
This configuration evaluates a mass-optimization approach by replacing all components (tubes and brackets) with 6061-T6 Aluminum. While it significantly reduces the unsprung mass of the suspension, the lower elastic modulus ($E = 70 \text{ GPa}$) increases global deflection. This higher flexibility significantly lowers the structural Factor of Safety compared to the steel baseline. The shock absorber brackets were thickened to reach the desired FOS in those components, from 3 mm to 6.35 mm.

SetUp:

Fixed hinge on bushings rotation axis.

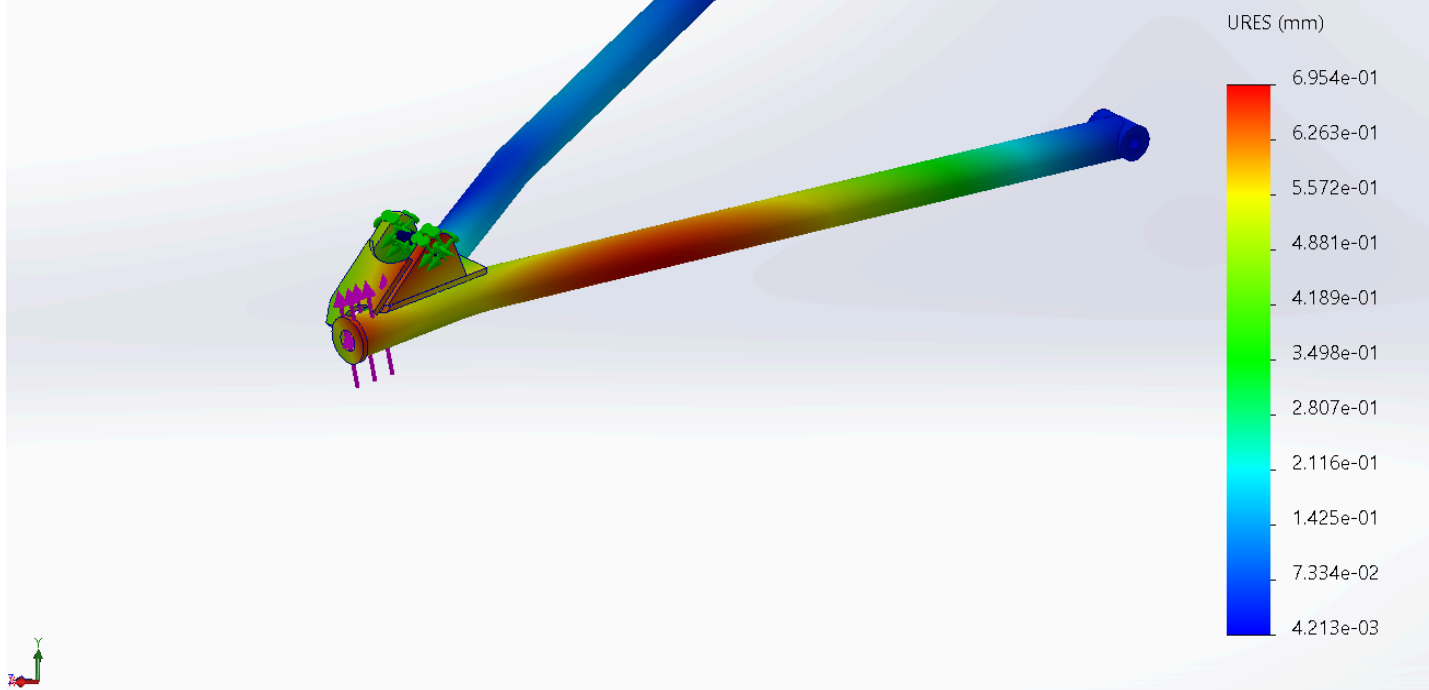
Hinge on shock absorber bearings limited translation through the shock absorber direction (full compression), with bolt connection between the two bearings.

Forces applied on the ball joint bearing connection.

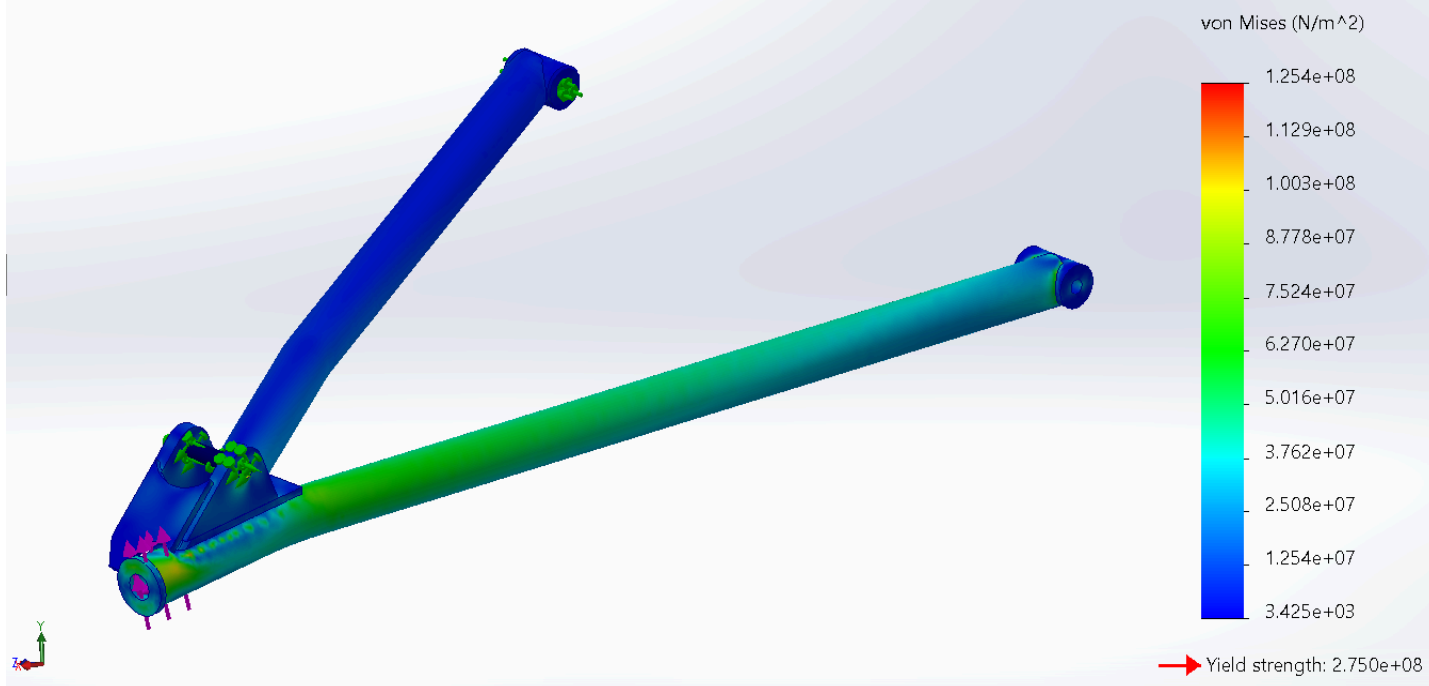


Results:

Model name: SUS_FT00_ASSMBLY FRONT TOP ARM
Study name: FinalVersion(-Predeterminado-)
Plot type: Static displacement Displacement1
Deformation scale: 1



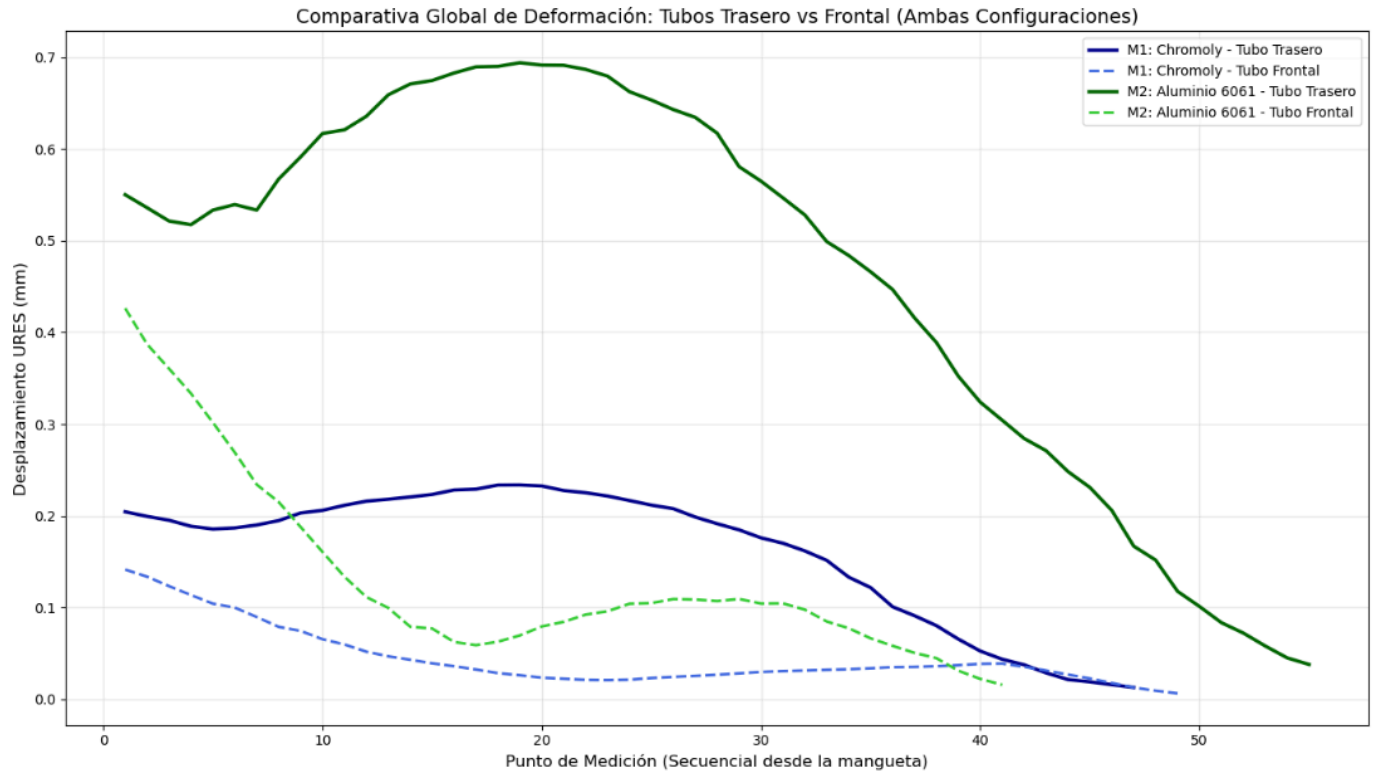
Model name: SUS_FT00_ASSMBLY FRONT TOP ARM
Study name: FinalVersion(-Predeterminado-)
Plot type: Static nodal stress (Top) Stress1



MATRIZ DE DECISIÓN ESTRUCTURAL						
	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Eficiencia_FOS_kg	Status
0	M1_4130_Tubos	110.1	4.18	1.87361	2.230987	Passed
1	M1_AISI1020_Orejas	162.3	2.16	1.87361	1.152855	Passed
2	M2_Aluminio6061_Tubos	106.8	2.57	0.64390	3.991303	Passed
3	M2_Aluminio6061_Orejas	125.4	2.19	0.64390	3.401149	Passed

MATRIZ DE DECISIÓN ESTRUCTURAL AVANZADA						
	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Rigidez_K_N_mm	Rigidez_Especifica
0	M1_Chromoly_Tubos	110.1	4.2	1.9	9005.5	4806.5
1	M2_Aluminio_Tubos	106.8	2.6	0.6	3075.7	4776.7

REPORTE FINAL DE OPTIMIZACIÓN Y ELASTOCINEMÁTICA				
	Performance Metric	Modelo 1 (Chromoly)	Modelo 2 (Aluminio)	Delta (%)
0	Total Mass (kg)	1.87361	0.6439	-65.6%
1	FOS Min [Tubos]	4.18000	2.5700	-38.5%
2	Global Stiffness K (N/mm)	9005.50000	3075.7000	-65.8%
3	Specific Stiffness ((N/mm)/kg)	4806.50000	4776.7000	-0.6%



Rear Upper A-Arms

Configuration 1:

This configuration represents the baseline design, utilizing an all-steel construction. The main tubular members are modeled in AISI 4130 Chromoly, as well as the brackets. This setup provides high global stiffness ($E = 200$ GPa) and excellent yield strength, ensuring minimal deflection and a high Factor of Safety (FOS) under Worst-Case Scenario (WCS) loads. However, it represents the heaviest iteration of the A-Arm.

Configuration 2:

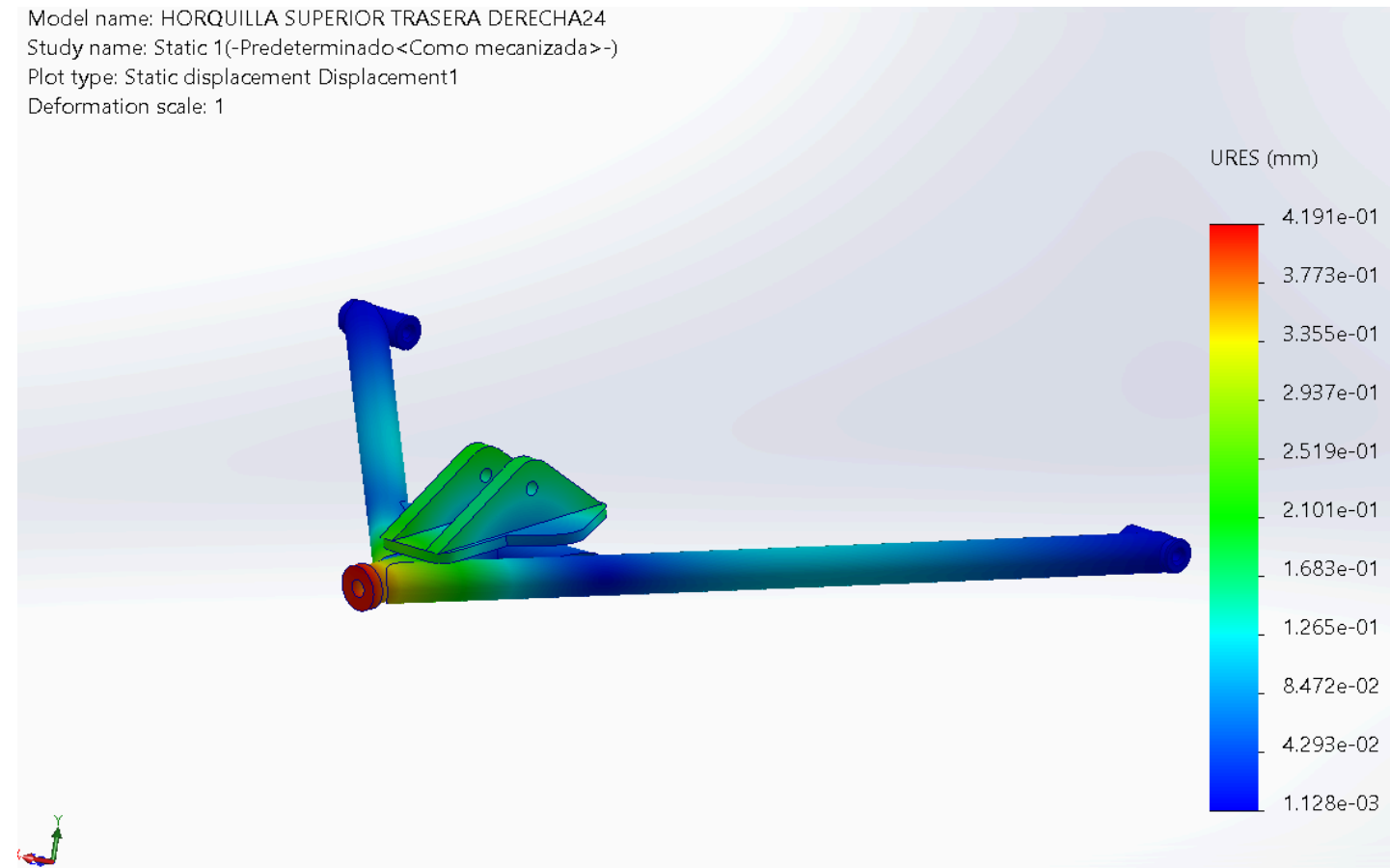
This configuration evaluates a mass-optimization approach by replacing all components (tubes and brackets) with 6061-T6 Aluminum. While it significantly reduces the unsprung mass of the suspension, the lower elastic modulus ($E = 70$ GPa) increases global deflection. This higher flexibility significantly lowers the structural Factor of Safety compared to the steel baseline. The shock absorber brackets were thickened to reach the desired FOS in those components, from 3 mm to 9.525 mm.

SetUp:

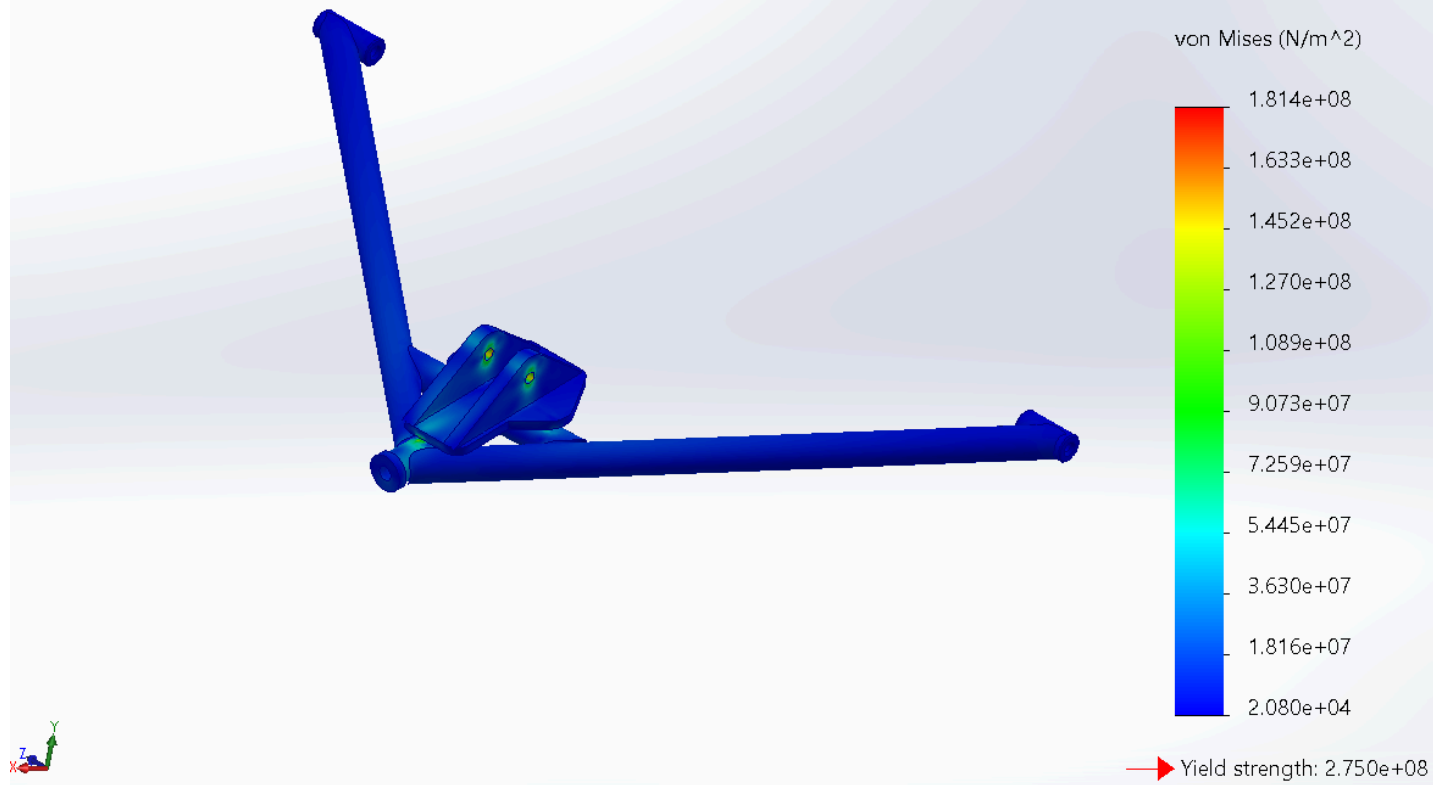
Fixed hinge on bushings rotation axis.

Hinge on shock absorber bearings limited translation through the shock absorber direction (full compression), with bolt connection between the two bearings.

Forces applied on the ball joint bearing connection.



Model name: HORQUILLA SUPERIOR TRASERA DERECHA24
Study name: Static 1(-Predeterminado<Como mecanizada>-)
Plot type: Static nodal stress (Top) Stress1
Deformation scale: 1



MATRIZ DE DECISIÓN ESTRUCTURAL

	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Eficiencia_FOS_kg	Status
0	M1_4130_Tubos	140.0	3.29	2.713	1.212680	Passed
1	M1_AISI1020_Orejas	109.8	4.19	2.713	1.544416	Passed
2	M2_Aluminio6061_Tubos	148.4	1.85	0.933	1.982851	Failed
3	M2_Aluminio6061_Orejas	181.4	1.52	0.933	1.629153	Failed

MATRIZ DE DECISIÓN ESTRUCTURAL AVANZADA

	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Rigidez_K_N_mm	Rigidez_Especificas
0	M1_Chromoly_Tubos	140.0	3.3	2.7	23669.9	8724.6
1	M2_Aluminio_Tubos	148.4	1.8	0.9	5935.5	6361.7

REPORTE FINAL DE OPTIMIZACIÓN Y ELASTOCINEMÁTICA

	Performance Metric	Modelo 1 (Chromoly)	Modelo 2 (Aluminio)	Delta (%)
0	Total Mass (kg)	2.713	0.933	-65.6%
1	FOS Min [Tubos]	3.290	1.850	-43.8%
2	Global Stiffness K (N/mm)	23669.900	5935.500	-74.9%
3	Specific Stiffness ((N/mm)/kg)	8724.600	6361.700	-27.1%

